PHASE - 2

1. Introduction

1.1 Project Overview

Avento is a centralized web-based event management platform designed to simplify and digitize the complete event management process within educational institutions. The platform provides a unified environment for students, faculty coordinators, student organizers, and administrators to manage and participate in academic, technical, cultural, and extracurricular events efficiently.

The system supports the complete event lifecycle, including event creation, approval, registration, attendance tracking, QR-code-based check-in, notifications, analytics, and certificate distribution. By consolidating these activities into a single platform, Avento eliminates the dependency on multiple disconnected tools such as notice boards, emails, messaging groups, spreadsheets, and paper-based registration methods.

Built using modern web technologies and cloud-based services, Avento aims to improve accessibility, transparency, operational efficiency, and student engagement while reducing the administrative effort involved in organizing institutional events.

1.2 Objectives

The primary objectives of Avento are:

- To provide a centralized platform for managing institutional events.
- To simplify event discovery and registration for students.
- To support both individual and team-based event participation.
- To streamline attendance tracking through QR-based verification and automate certificate generation and distribution.
- To enable efficient communication through notifications and updates.
- To provide role-based access and event management capabilities for organizers and administrators.
- To generate analytical insights that support informed decision-making and event planning.

1.3 Scope of the System

Avento is intended for use within educational institutions to manage academic, technical, cultural, and extracurricular events. The platform supports multiple user roles, including Super Admin, Faculty Coordinator, Student Organizer, and Student.

The system covers event proposal submission, approval workflows, event creation, registration management, attendance monitoring, QR-based check-in, certificate generation, notification services, analytics, and audit logging. Future enhancements may include AI-powered recommendations, intelligent search capabilities, waitlist management, and mobile application support.

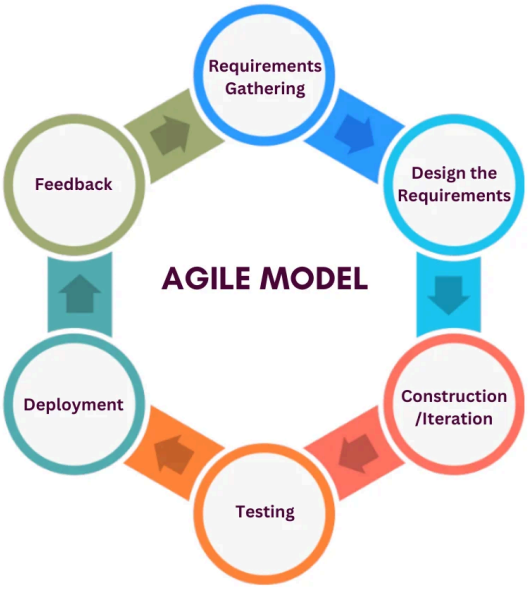
2. Software Development Methodology

2.1 Selected SDLC Model

Agile Incremental Model

The Agile Incremental Model has been selected for the development of Avento. The project consists of multiple independent modules such as authentication, event management, registration, attendance tracking, notifications, analytics, and certificate generation. Therefore, the system can be developed and tested incrementally while continuously incorporating feedback and improvements.

Development Flow



2.2 Justification

Project Requirement	Reason for Choosing Agile Incremental Model
Multiple Modules	Each module such as event management, registration, attendance, certificates, and analytics can be developed independently.

Continuous Review	Feedback from students, organizers, and administrators can be incorporated after every iteration.
Changing Requirements	New features and modifications can be added without redesigning the entire system.
Better Testing	Individual modules can be tested and validated before integration with other components.
Scalability	Future enhancements such as AI recommendations, mobile applications, and advanced analytics can be added easily.
Reduced Risk	Development issues and defects can be identified and resolved early in each increment.
Faster Delivery	Core functionalities can be released and evaluated before implementing advanced features.
User Involvement	Stakeholders can actively participate in reviewing increments, ensuring the system meets institutional requirements.

2.3 Development Phases

Phase	Activities
Requirement Analysis	Problem identification, stakeholder analysis, functional and non-functional requirement gathering.
System Design	Architecture design, database design, workflow design, ERD preparation, and DFD preparation.
Development	Implementation of frontend, backend services, database, and business logic.
Testing	Unit testing, integration testing, system testing, and user acceptance testing.
Deployment	System deployment, configuration, and user onboarding.
Maintenance & Enhancement	Bug fixes, performance optimization, feature enhancements, and future updates.

3. Requirement Analysis

This phase focuses on identifying the needs of all stakeholders involved in the event management process, including students, faculty coordinators, student organizers, and

administrators. The requirements were gathered through analysis of existing event management practices, institutional workflows, and identified challenges in event registration, attendance tracking, communication, and certificate distribution.

The detailed functional requirements of the system are documented separately in the Functional Requirements Specification.

Non-Functional Requirements:

Category	Requirement
Performance	The system should provide fast response times during event registration, attendance verification, and user interactions.
Availability	The platform should remain accessible to users except during scheduled maintenance.
Security	Authentication, role-based authorization, OTP verification, and data protection mechanisms shall be implemented to ensure secure access.
Reliability	The system should accurately maintain event, registration, attendance, and certificate records.
Scalability	The platform should support increasing numbers of users, events, and registrations.
Usability	The user interface should be simple, intuitive, and easy to navigate.
Maintainability	The system should be modular and easy to update or enhance.
Auditability	Critical actions should be logged for monitoring and accountability.
Compatibility	The platform should function across modern web browsers and responsive devices, including desktops, tablets, and smartphones.
Data Integrity	Stored information should remain accurate, consistent, and protected from unauthorized modifications.

4. System Architecture & Technology Stack

4.1 System Architecture

Avento is designed as a centralized event management platform that connects students, organizers, faculty coordinators, and administrators through a unified digital environment. The system follows a modular architecture where each functional component operates independently while remaining integrated with the overall platform.

The architecture supports the complete event lifecycle, beginning with event creation and approval, followed by participant registration, attendance management, communication, certificate distribution, and performance analysis. This modular approach improves maintainability, simplifies future enhancements, and ensures efficient handling of institutional events.

The platform is designed to provide secure access, real-time information management, and seamless interaction between different user roles while maintaining data consistency across all modules.

4.2 Technology Stack

To support the requirements of a modern event management platform, Avento utilizes a combination of web technologies, cloud services, and database solutions. The selected technologies provide responsive user experiences, secure authentication, efficient data management, and scalable system performance.

The user interface is developed using modern frontend technologies to ensure accessibility across different devices. Backend services handle authentication, business logic, and communication with the database. A relational database management system is used to store and manage user information, event records, registrations, attendance data, certificates, and system logs.

The technology stack has been selected based on factors such as scalability, maintainability, security, ease of development, and support for future enhancements.

Technology	Purpose
React.js	Frontend development and user interface creation
Bootstrap	Responsive design and UI styling
Supabase	Authentication, backend services, and cloud integration
PostgreSQL	Relational database management and data storage

4.3 Module Overview

The Avento platform is organized into several functional modules that collectively support the management of institutional events.

- Authentication Module
- Event Management Module
- Registration & Team Management Module
- Attendance Management Module
- Certificate Management Module

- Notification Module
- Analytics & Reporting Module
- Audit Log Module

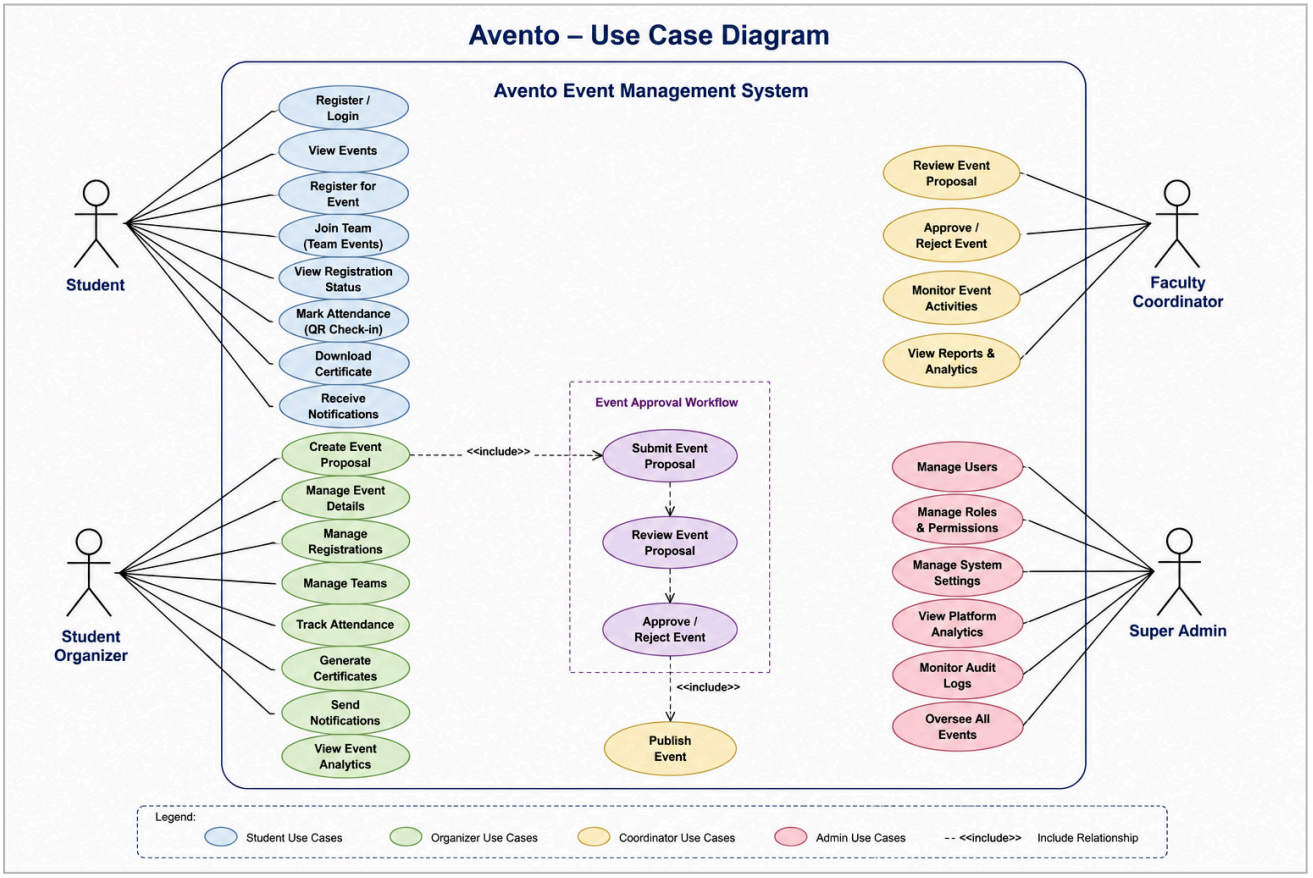
Each module is designed to perform a specific set of responsibilities while interacting with other modules to provide a seamless event management experience.

5. Software Modeling

Software modeling provides a visual representation of the system's structure, functionality, and interactions. It helps in understanding the behavior of the system before implementation and serves as a blueprint for development.

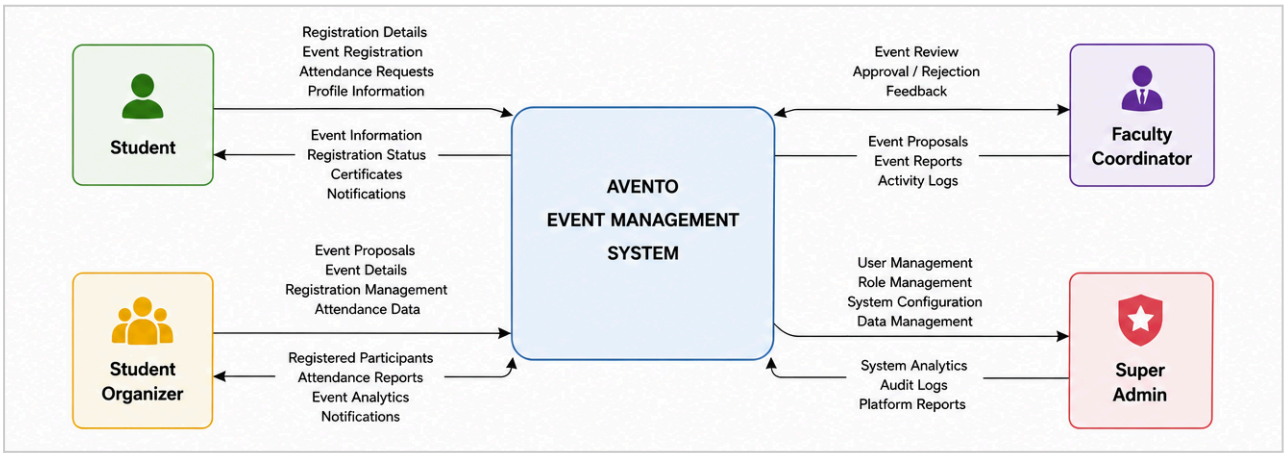
For the Avento platform, software modeling techniques such as Use Case Diagrams, Class Diagrams, Entity Relationship Diagrams (ERD), and Data Flow Diagrams (DFD) are used to represent user interactions, system components, data relationships, and information flow. These models assist in analyzing requirements, designing system architecture, and ensuring that the developed system meets stakeholder expectations.

5.1 Use Case Diagram

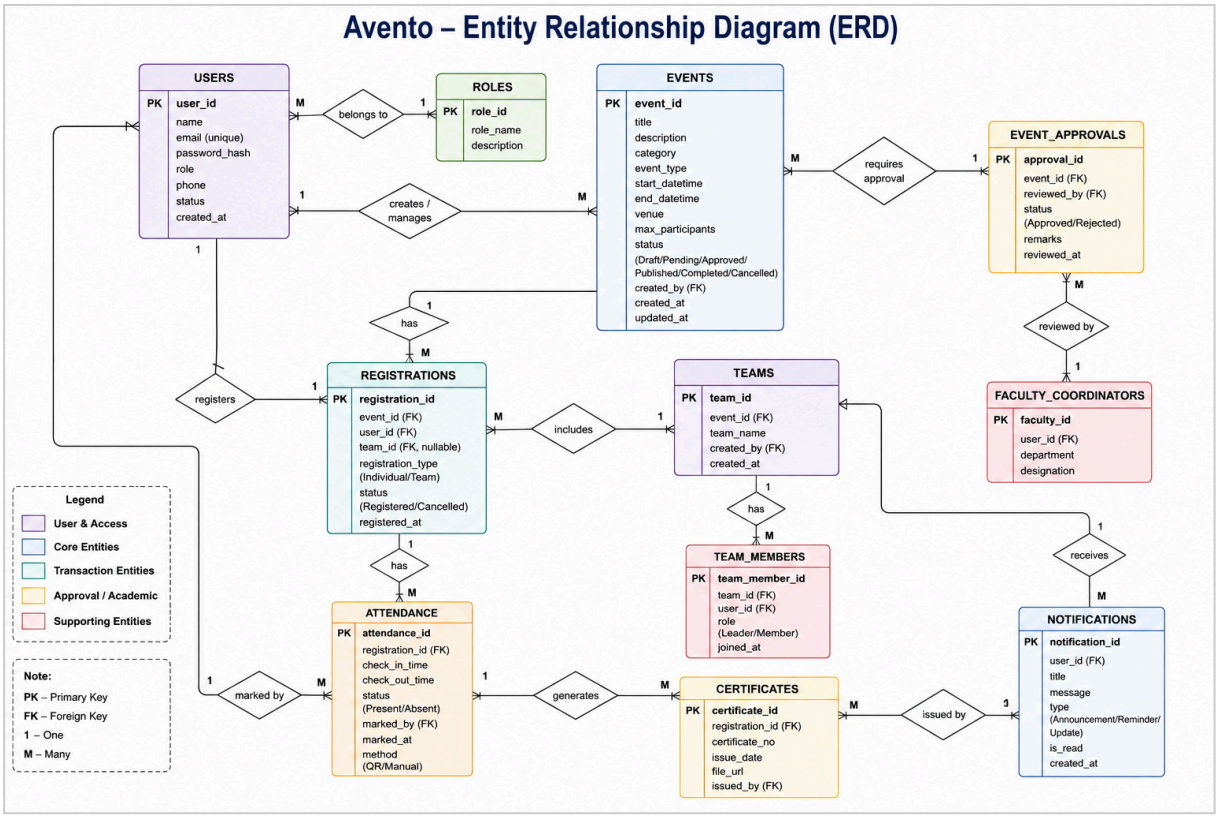


5.2 Data Flow Diagram (DFD)

Level 0



5.3 Entity Relationship Diagram (ERD)



6. Security Design

The security design of Avento focuses on protecting user accounts, event information, and institutional data from unauthorized access. Multiple security mechanisms are implemented to ensure confidentiality, integrity, and controlled access to system resources.

OTP Authentication:
Avento uses One-Time Password (OTP) verification during user registration and authentication processes. This ensures that only users with valid institutional email addresses can access the platform and helps prevent unauthorized account creation.

Role-Based Access Control (RBAC):
The platform implements role-based access control to regulate user permissions. Different roles, including Student, Student Organizer, Faculty Coordinator, and Super Admin, are assigned specific privileges based on their responsibilities within the system.

Session Management:
Secure session management mechanisms are used to maintain authenticated user sessions and prevent unauthorized access. Sessions are terminated upon logout, and protected resources can only be accessed by authenticated users.

Database Security Policies:
Database-level security policies are implemented to protect sensitive information and restrict direct access to data. Access permissions are enforced to ensure that users can only view and modify data relevant to their assigned roles.

7. Testing Strategy

Testing is performed throughout the development lifecycle to ensure that the Avento platform functions correctly, securely, and reliably. Different testing approaches are used to validate individual components as well as the complete integrated system.

Unit Testing:
Individual modules and functions are tested independently to verify that each component performs its intended operation correctly.

Integration Testing:
Integration testing ensures that multiple modules, such as authentication, event management, registration, attendance, and certificate generation, interact properly with one another.

System Testing:
The complete system is tested in an environment that closely resembles real-world usage. This testing validates overall functionality, performance, security, and usability.

User Acceptance Testing (UAT):
User Acceptance Testing is conducted with representative users to verify that the system

satisfies stakeholder requirements and supports institutional event management processes effectively.

8. Future Enhancements

Although Avento currently provides a comprehensive event management solution, several enhancements can further improve functionality, user experience, and system intelligence in future versions.

AI Event Recommendations and Smart Search:
Artificial intelligence can be used to recommend relevant events based on user interests, participation history, and preferences. Advanced search capabilities can also improve event discovery and information retrieval.

FAQ Chatbot and Mobile Application:
A chatbot can assist users by answering common queries related to registrations, schedules, and event participation. Additionally, a dedicated mobile application can provide greater accessibility and real-time engagement for students and organizers.

9. Conclusion

Avento provides a centralized and efficient platform for managing institutional events throughout their complete lifecycle. The system streamlines event creation, approval, registration, attendance tracking, notifications, analytics, and certificate distribution within a single integrated environment.

By adopting a modular architecture and Agile Incremental development approach, the platform ensures scalability, maintainability, and ease of future enhancement. The use of modern technologies and secure access mechanisms further contributes to the reliability and effectiveness of the system.

Overall, Avento addresses many of the challenges associated with traditional event management processes and offers a structured, transparent, and user-friendly solution for educational institutions.